Yandex Cloud

PG BootCamp Russia 2026 Moscow

PGBootCamp.ru

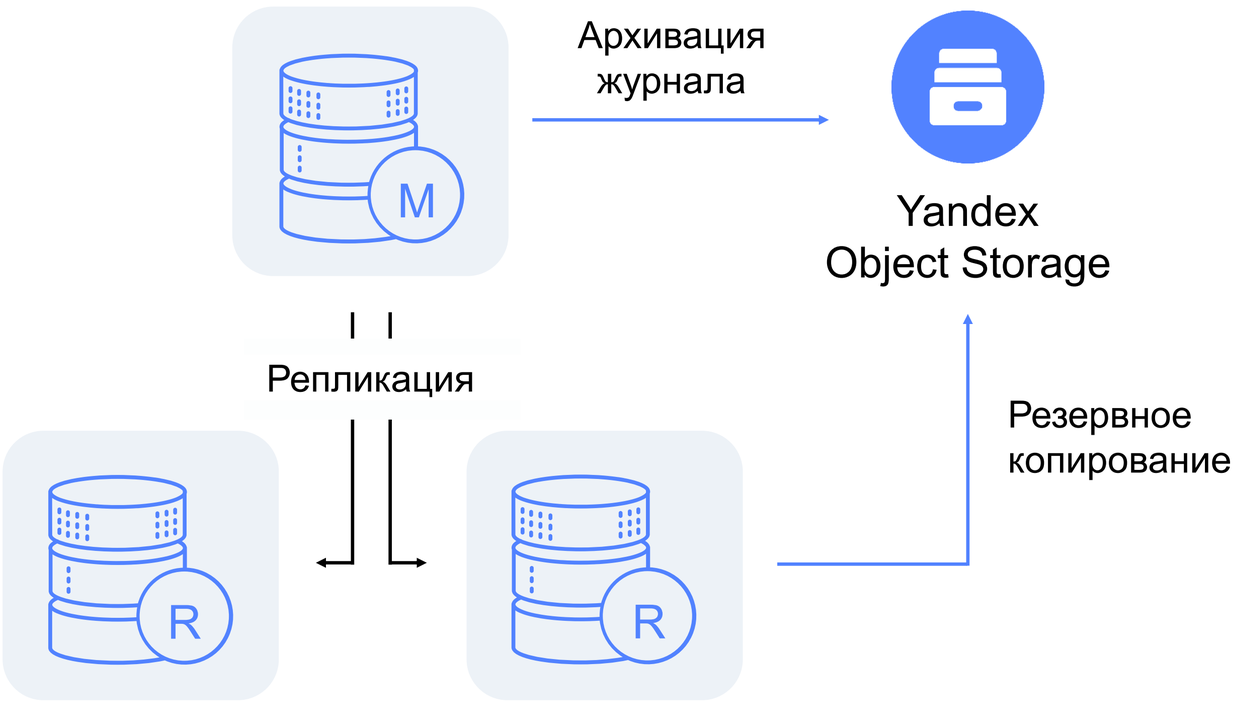
Архив будущего

Андрей Бородин, PostgreSQL hacker.

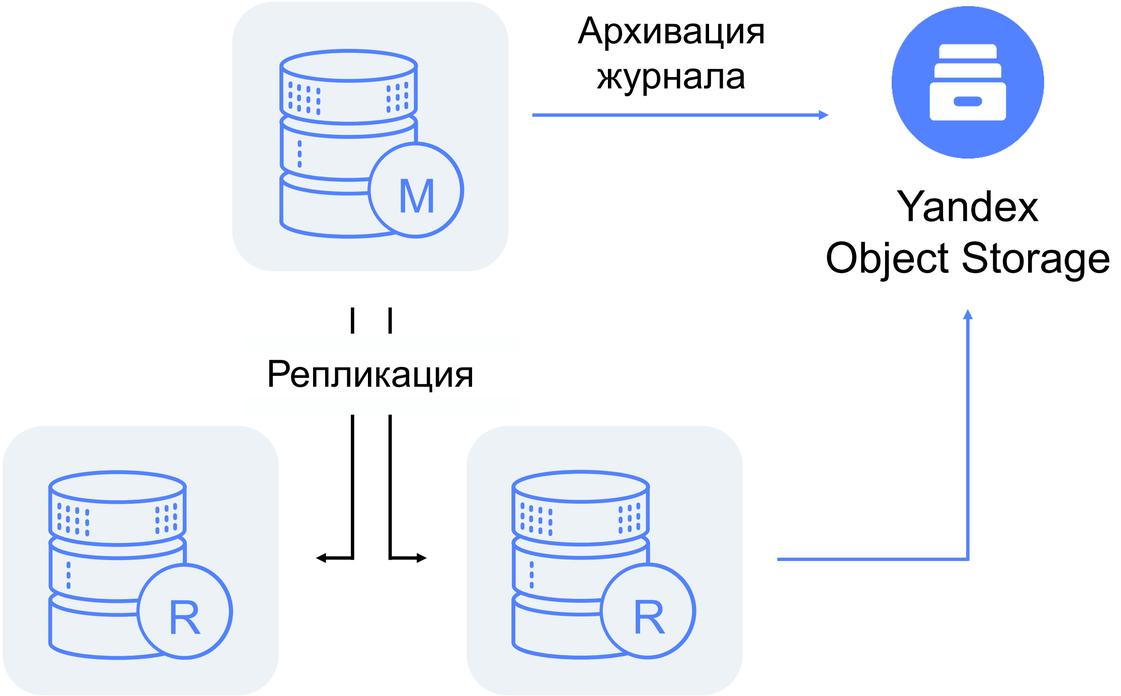
```
postgres % git log |grep Borodin|grep Andr| wc -l
     159
```

Open Source
С-ТРЕК
HighLoad++
Кирилл
Гаврилов
Яндекс
Кирилл
Решко
Yandex Cloud
радущийся
GOфер
Денис
Волков
Yandex Cloud
serverless
Yandex Cloud
FOOL CROSSING
DETOUR
DANGER
LUMPS AHEAD
TOM and JERRY

Топология HA-кластера в облаке

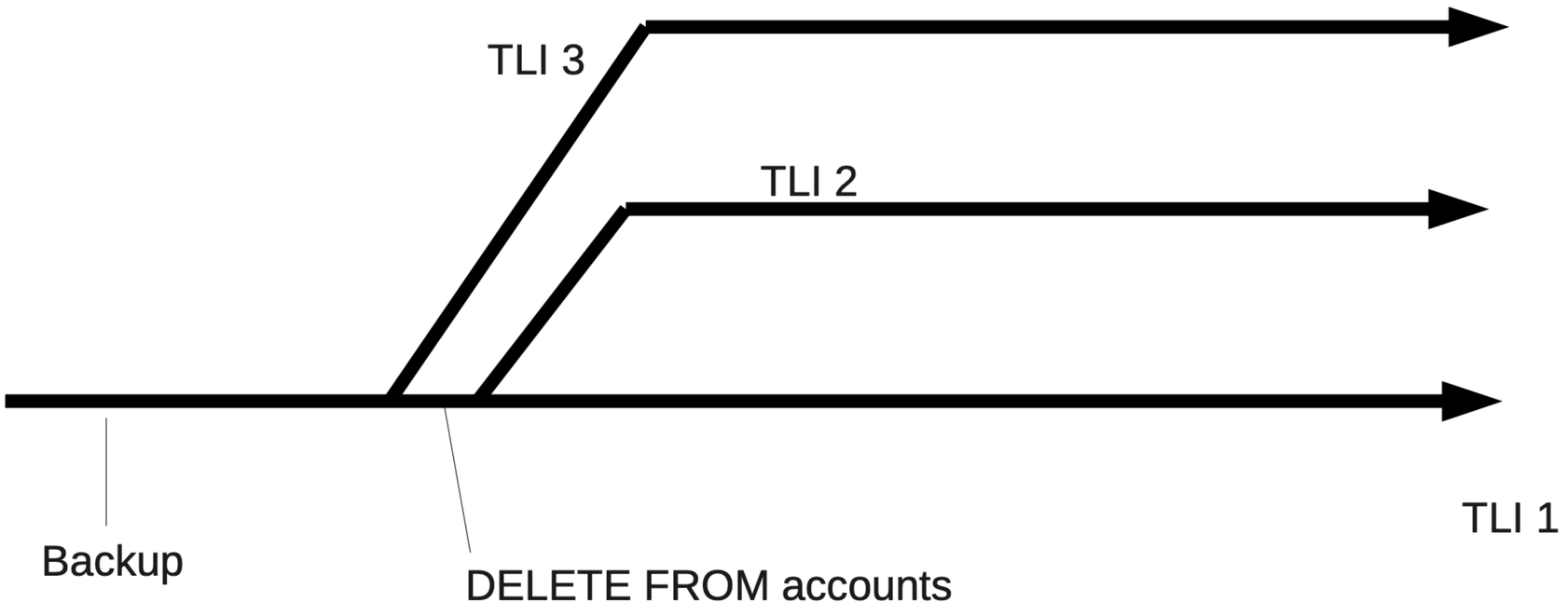


- Сегменты WAL
- Физическая
- И логическая репликация
- Модули архивации
- Timeline-ы



Timelines

- When new WAL is genereated after PITR, you don't want to overwrite the old WAL.



https://wiki.postgresql.org/images/e/e5/FOSDEM2013-Timelines.pdf

Проблемы других баз

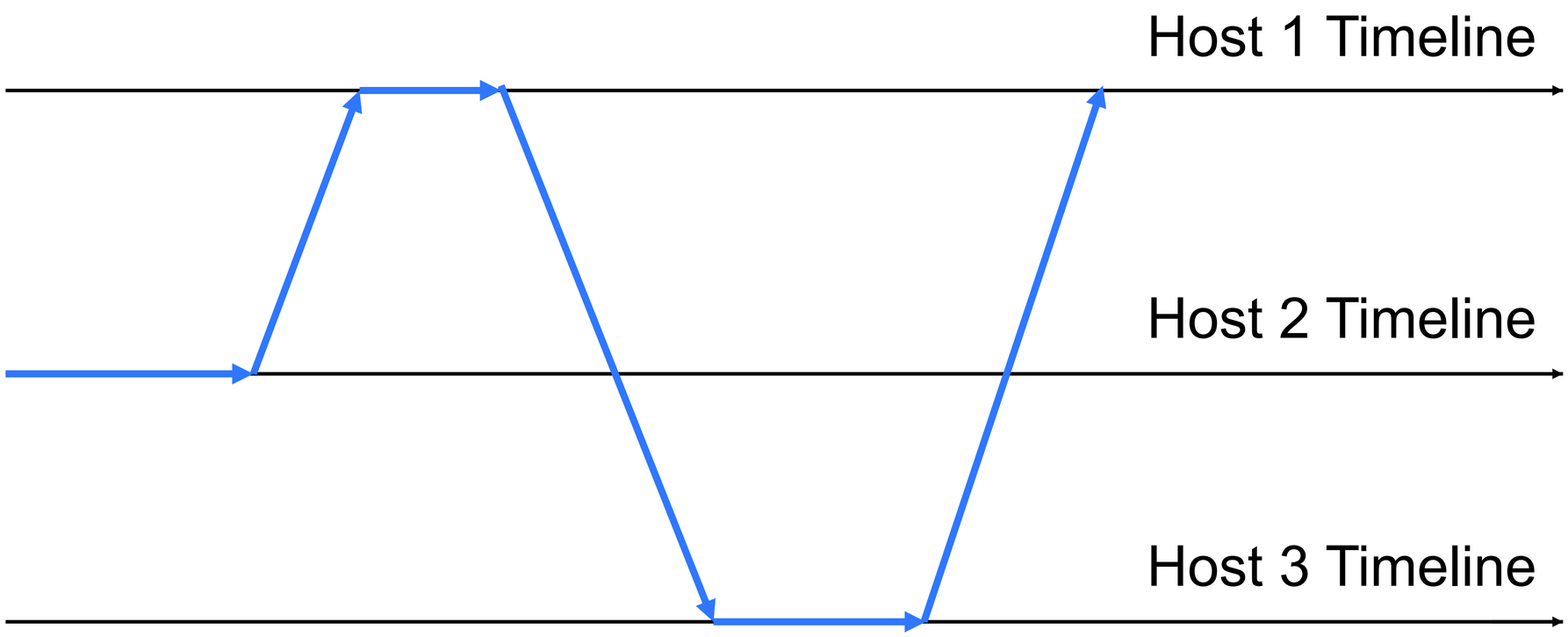
mongoDB®

MySQL: свои бин-логи на хостах

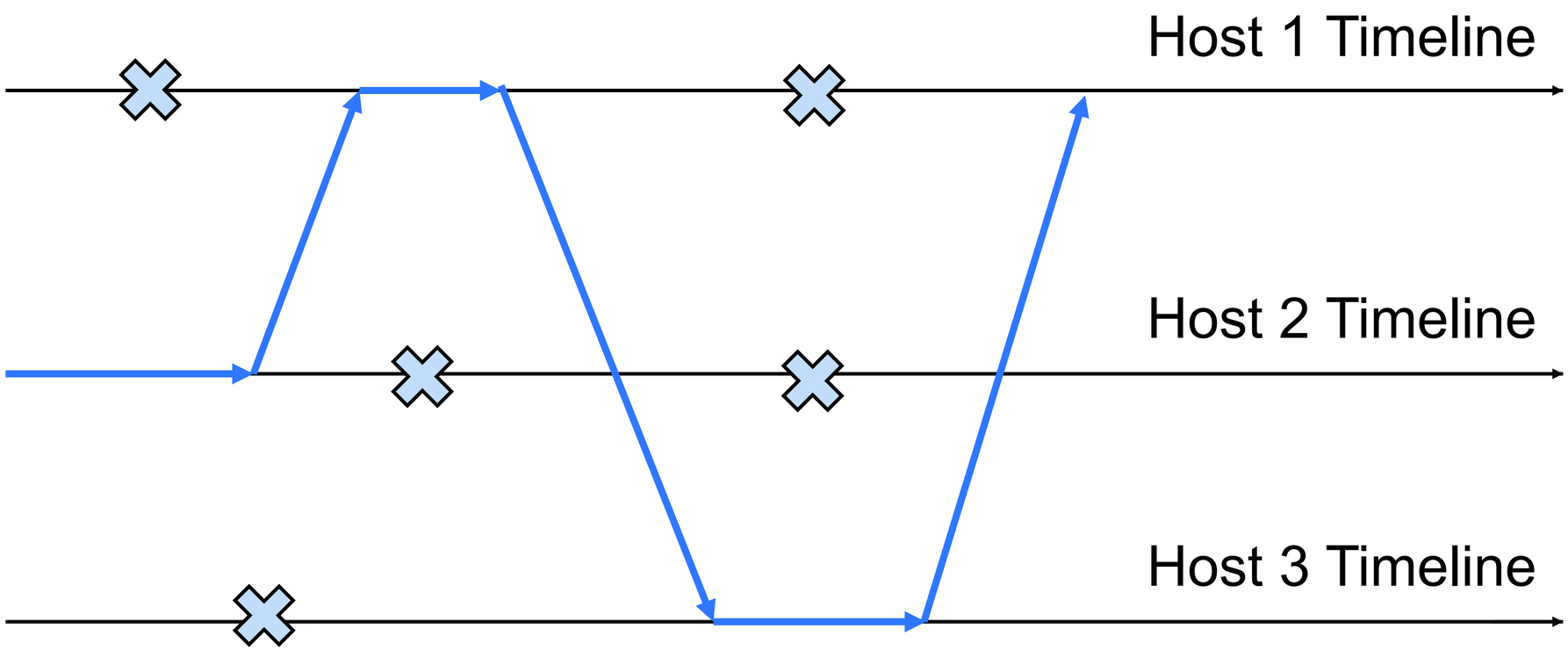
```
mysql> show binary logs;
+-------------------------------------------------------------+
| Log_name                                                    |
+-------------------------------------------------------------+
| mysql-bin-log-[illegible].000256                            |
| mysql-bin-log-[illegible].000257                            |
| mysql-bin-log-[illegible].000258                            |
```

```
mysql> show binary logs;
+-------------------------------------------------------------+
| Log_name                                                    |
+-------------------------------------------------------------+
| mysql-bin-log-[illegible].000249                            |
| mysql-bin-log-[illegible].000250                            |
| mysql-bin-log-[illegible].000251                            |
```

Switchover/Failover



Нелинейная история



Другие проблемы MySQL

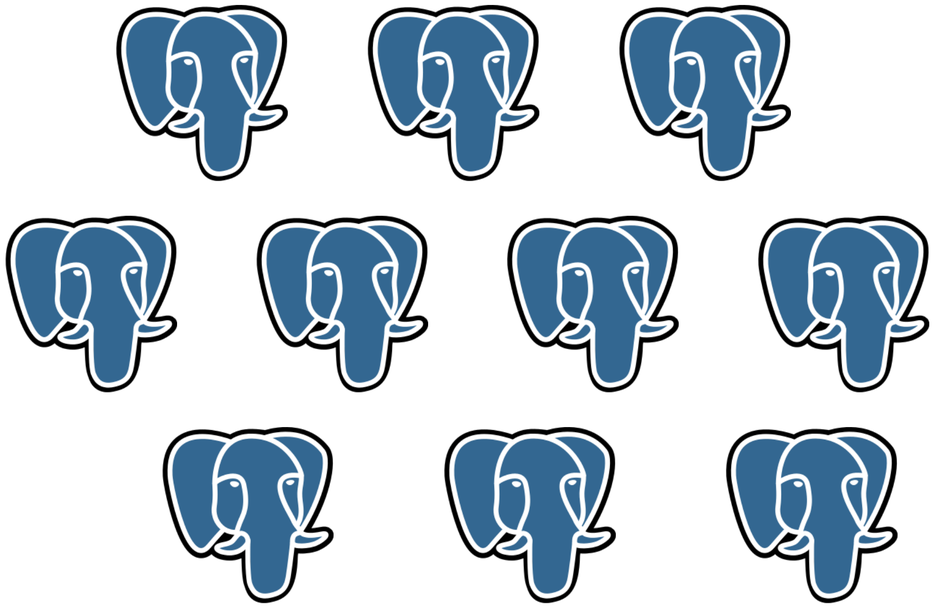
- Нет слотов репликации
- Нет “официального” механизма взаимодействия с архивом (archive_command), каждый пишет свой binlog server

GREENPLUM DATABASE®

Greenplum —
это же просто
много Postgres-ов!

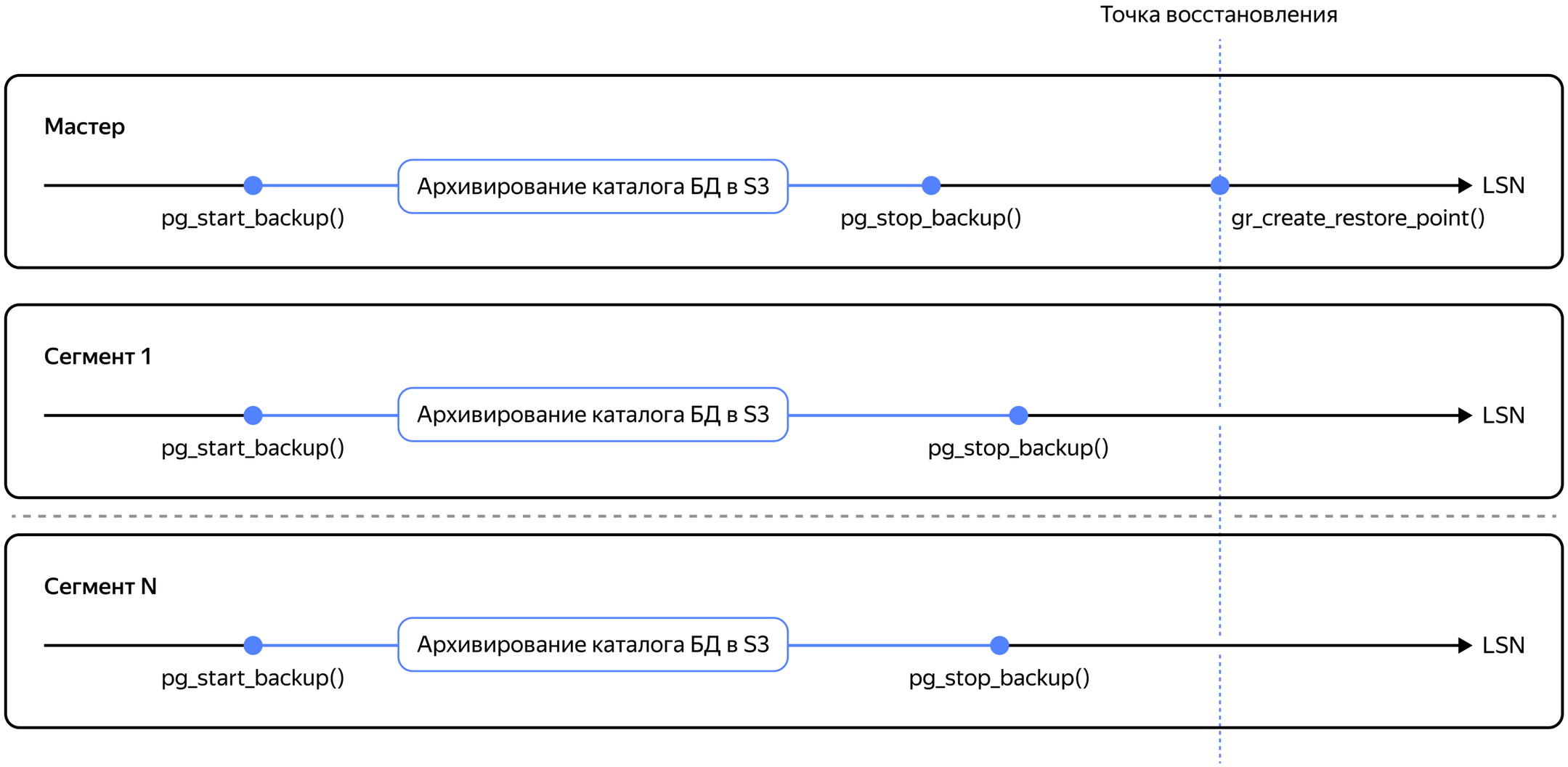
APACHE
Cloudberry

Apache Cloudberry —
это же просто
много Postgres-ов!



Apache Cloudberry — это же просто много Postgres-ов!

Физический бэкап Greenplum



Чего нет?
Реалистичные фичи

Флэшбеки

```
INSERT INTO some_table
( SELECT * FROM some_table
AS OF TIMESTAMP
TO_TIMESTAMP('2004-04-04 09:30:00', 'YYYY-MM-DD HH:MI:SS’)
WHERE id = 31337 );
```

В бекапе много лишнего

- Индексы
- FSM
- VM

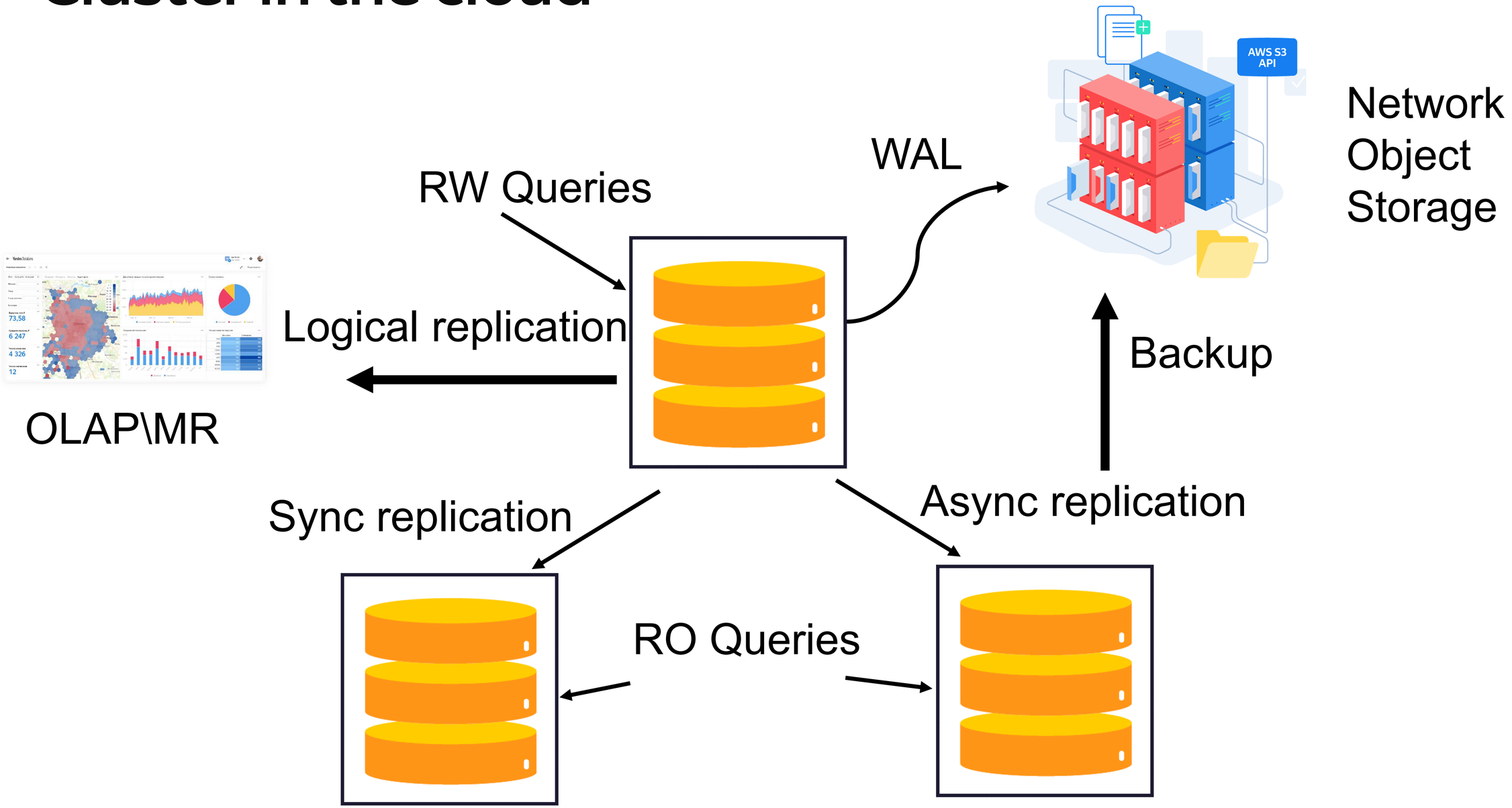
В резервной копии много лишнего

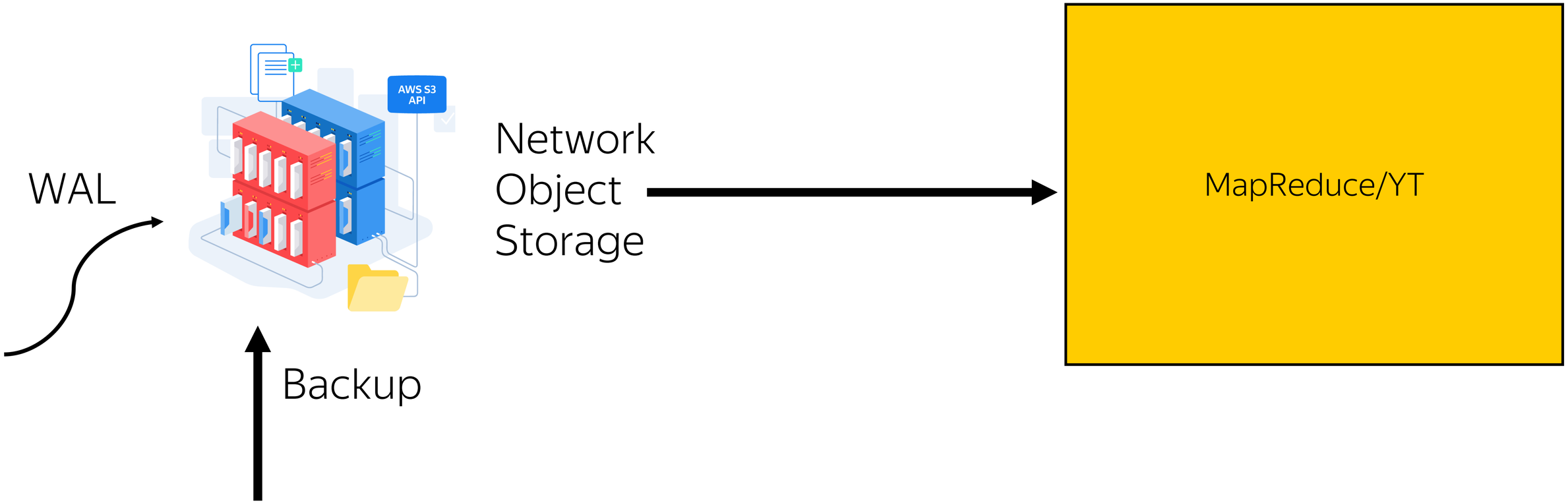
- Индексы
- FSM
- VM

Да и вообще бекап надо бы совместить с вакуумом

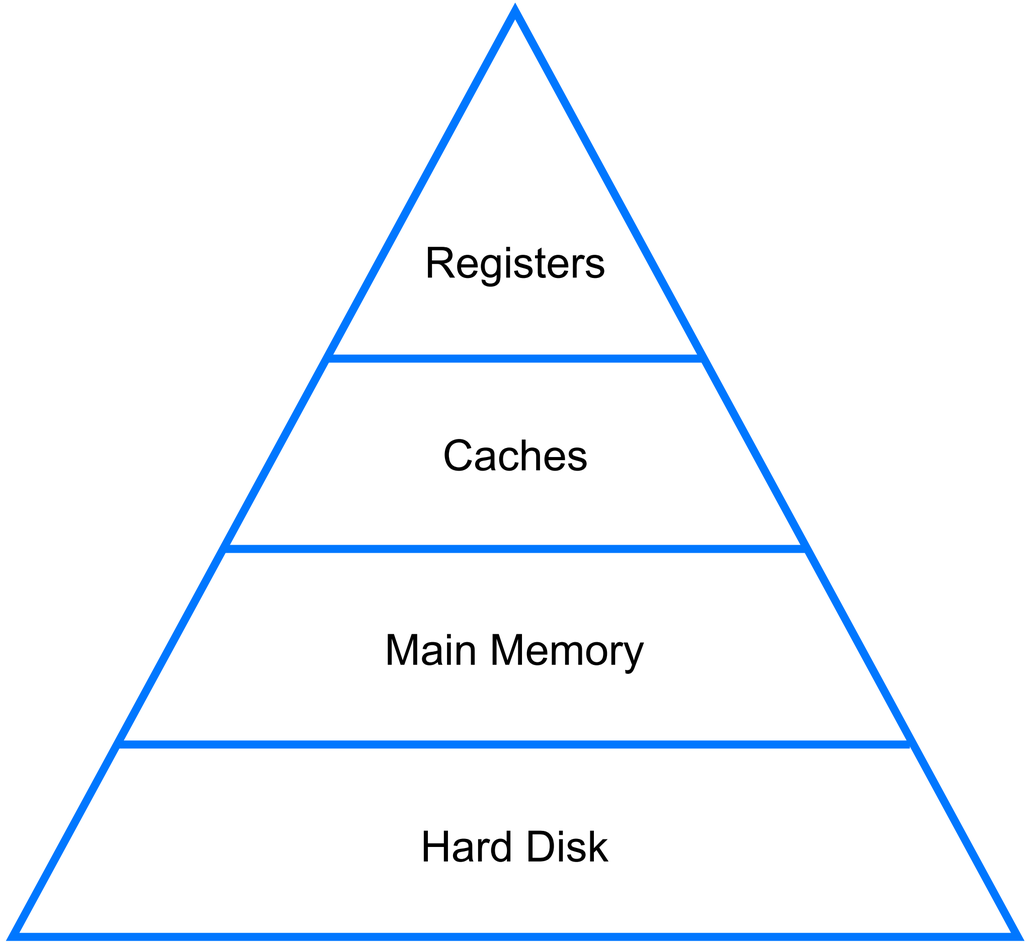
Рокетсаенс

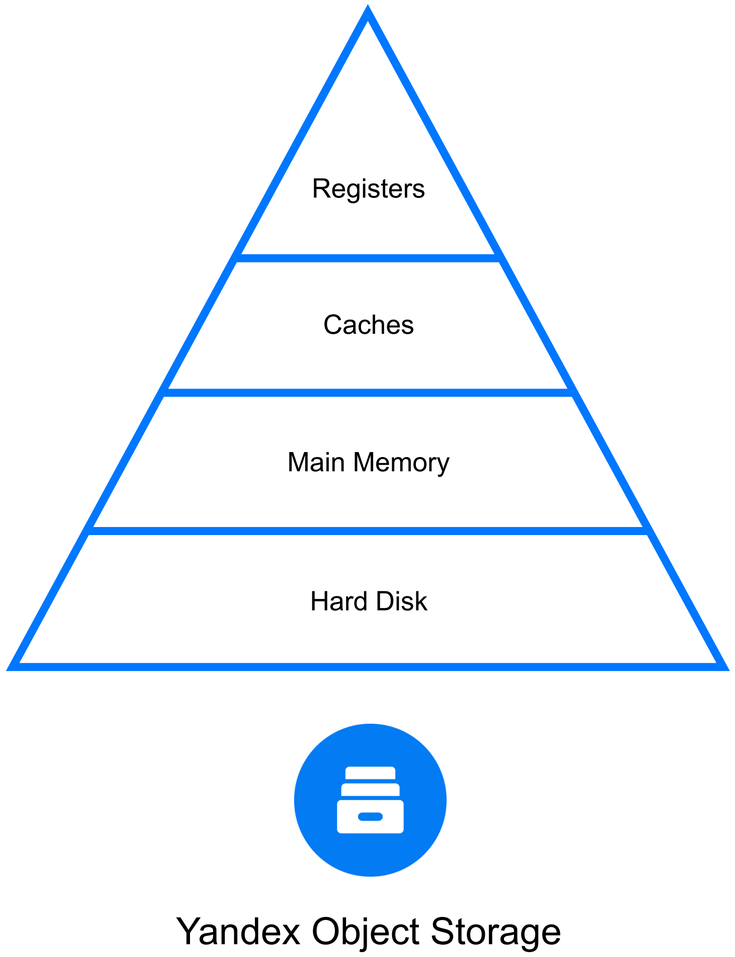
Cluster in the cloud



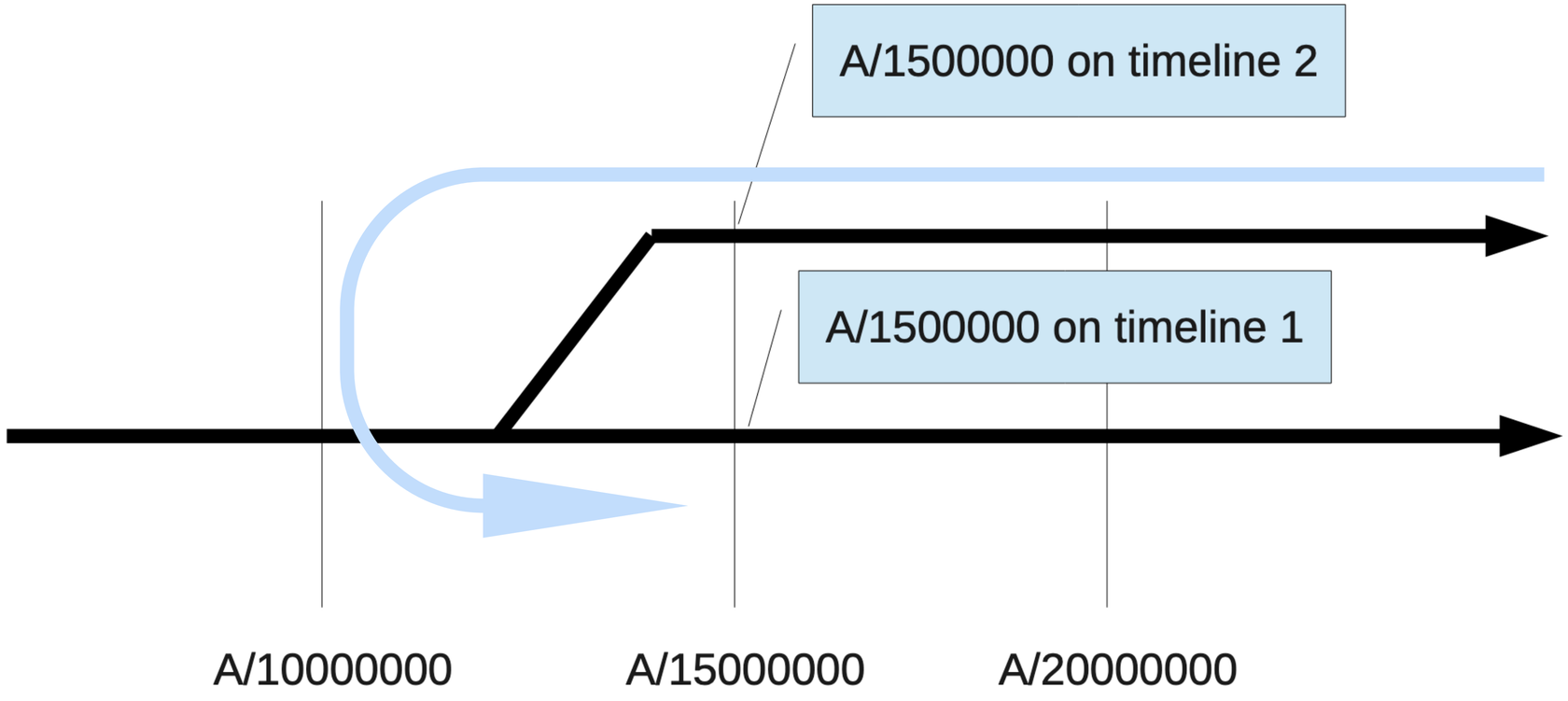

WAL
AWS S3
API
Network
Object
Storage
MapReduce/YT
Backup

Охлаждение лишних файлов в S3





Flashback vs Rewind



Над чем я работаю прямо сейчас

Shared archive

Home / Commitfest PG19-Final / Add archive_mode to prevent unarchived WAL on permanent primary failure

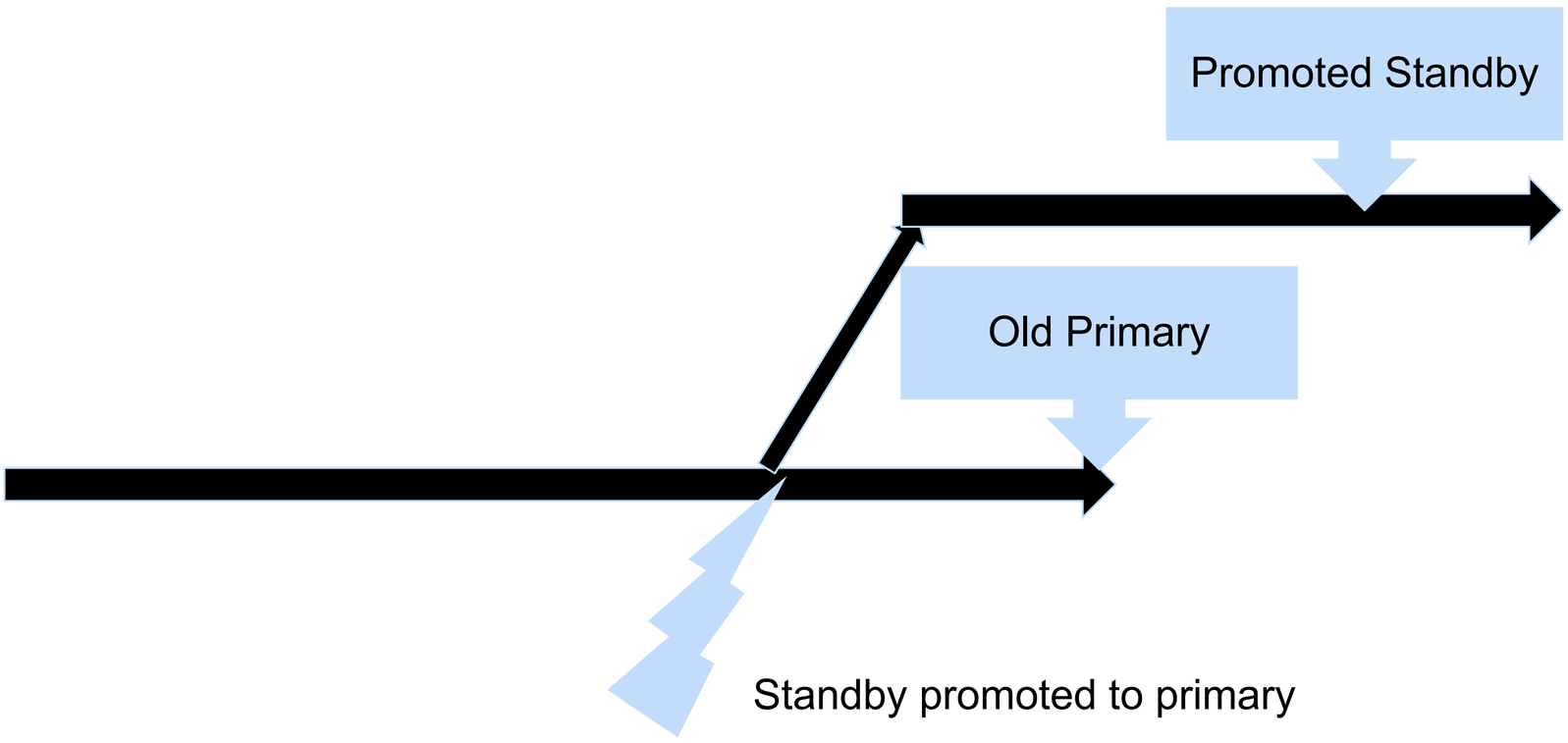
Add archive_mode to prevent unarchived WAL on permanent primary failure

Edit | Comment/Review | Change Status | Send private mail

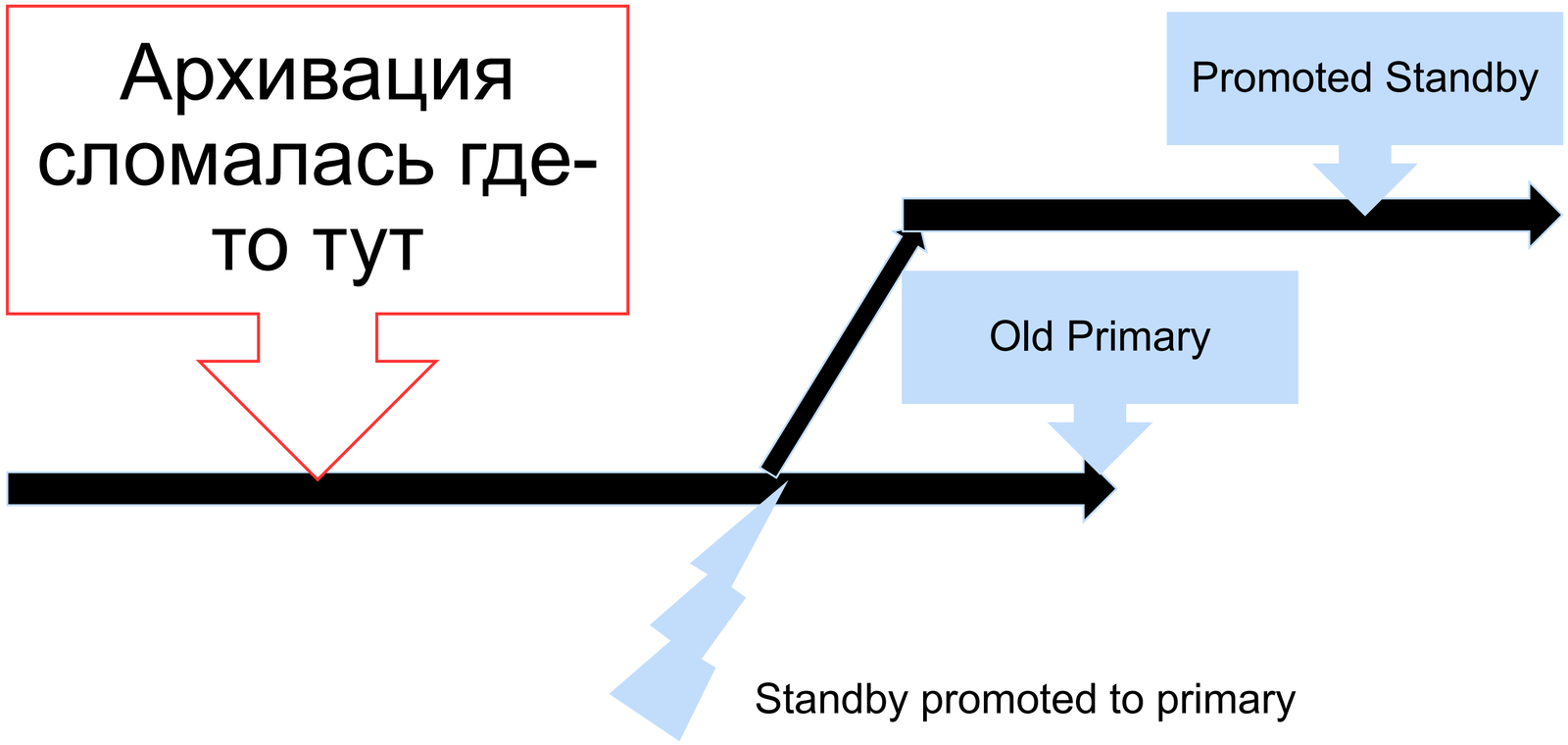
ID	6177
Title	Add archive_mode to prevent unarchived WAL on permanent primary failure
CI (CFBot)	Summary ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✗ ✓ Copy git checkout commands Requeue CFBot
Stats (from CFBot)	Patch version: v4, Patch count: 3, First patch: +599−38, All patches: +712−38
Tags	Physical Replication
Created	2025-10-29 08:16:23
Last modified	2026-02-13 06:55:30 (1 month ago)
Latest email	2026-03-16 09:43:15 (20 hours ago)
Status	PG19-Final (2026-03-01 – 2026-03-31): Needs review PG19-4 (2026-01-01 – 2026-01-31): Moved to different CF PG19-3 (2025-11-01 – 2025-11-30): Moved to different CF
Target version	19
Authors	Andrey Borodin (x4m)
Reviewers	

https://commitfest.postgresql.org/patch/6177/

Failover

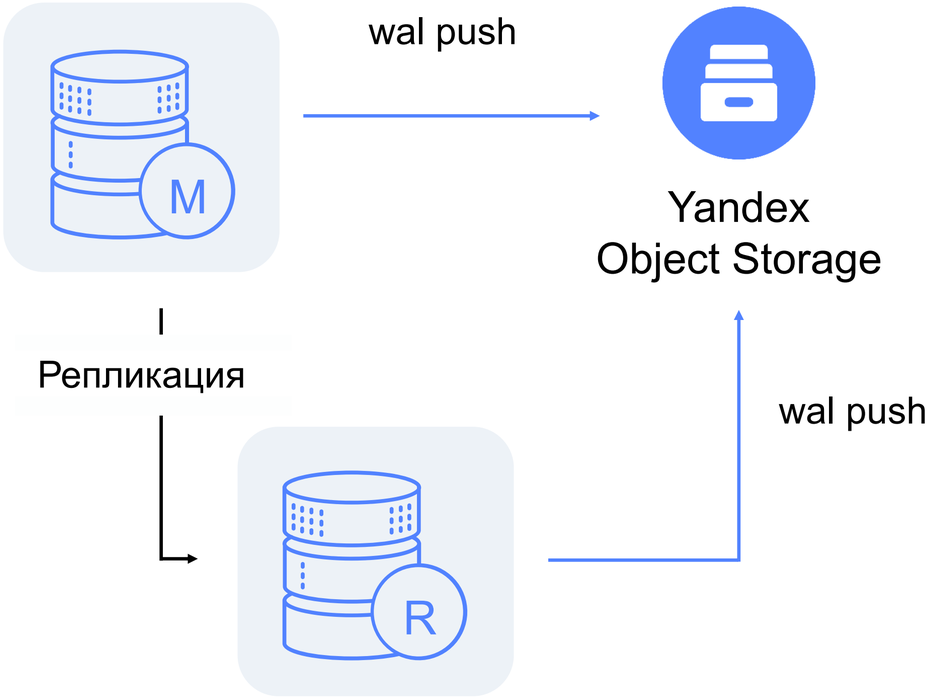


Failover


Архивация сломалась где-то тут
Promoted Standby
Old Primary
Standby promoted to primary

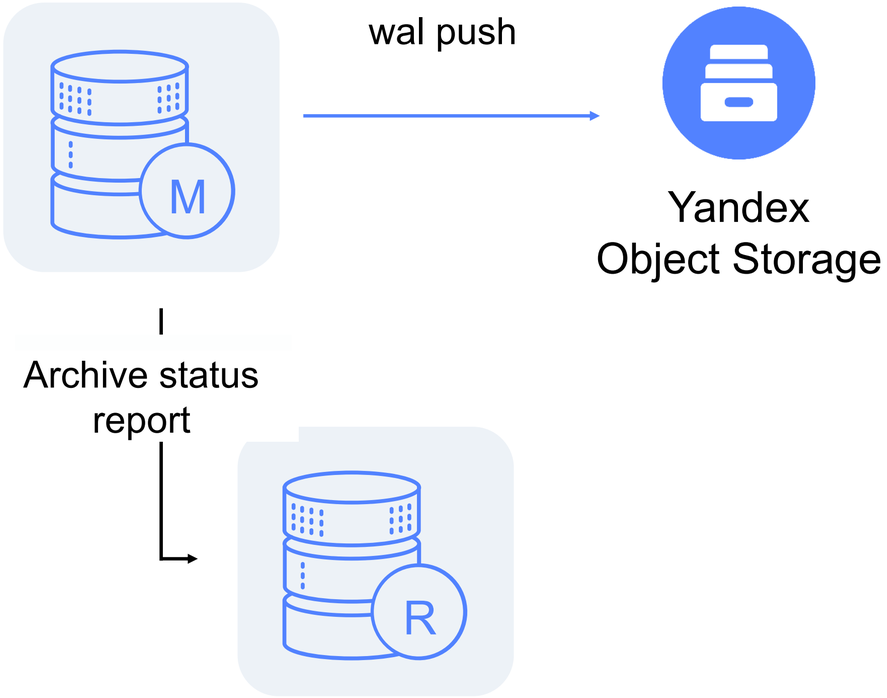
Возможное решение

archive_mode = always



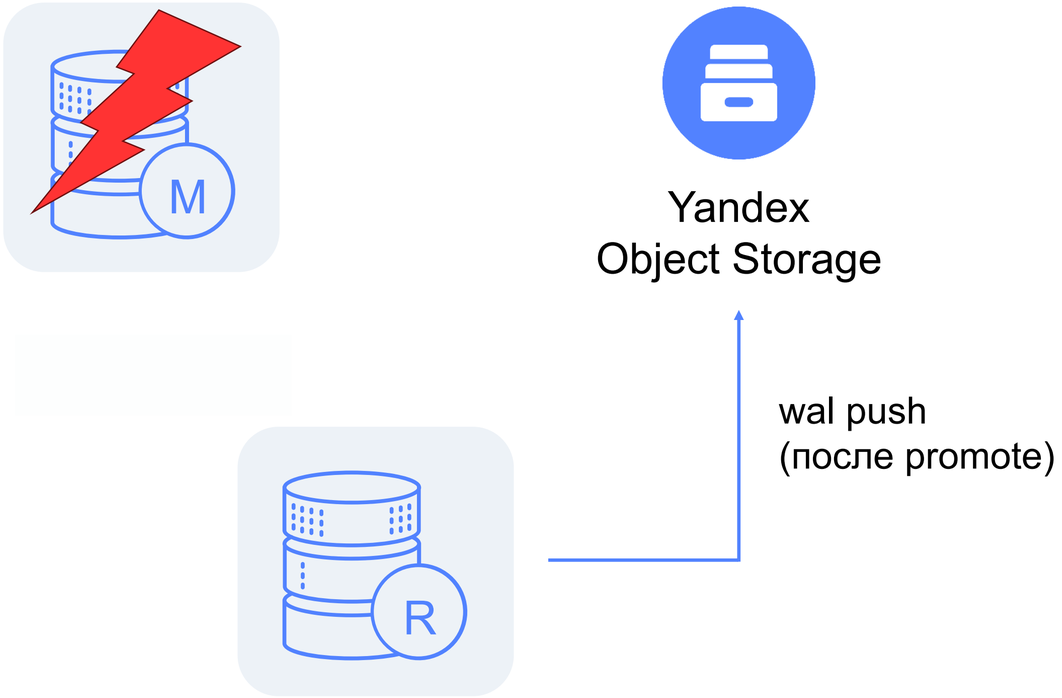
Правильное решение

archive_mode = shared_archive



Правильное решение

archive_mode = shared_archive



Timeline following

Home / Commitfest PG19-Final / Fix XLogFileReadAnyTLI silently applying divergent WAL from wrong timeline

Logged in as x4m (edit profile | log out | administration)

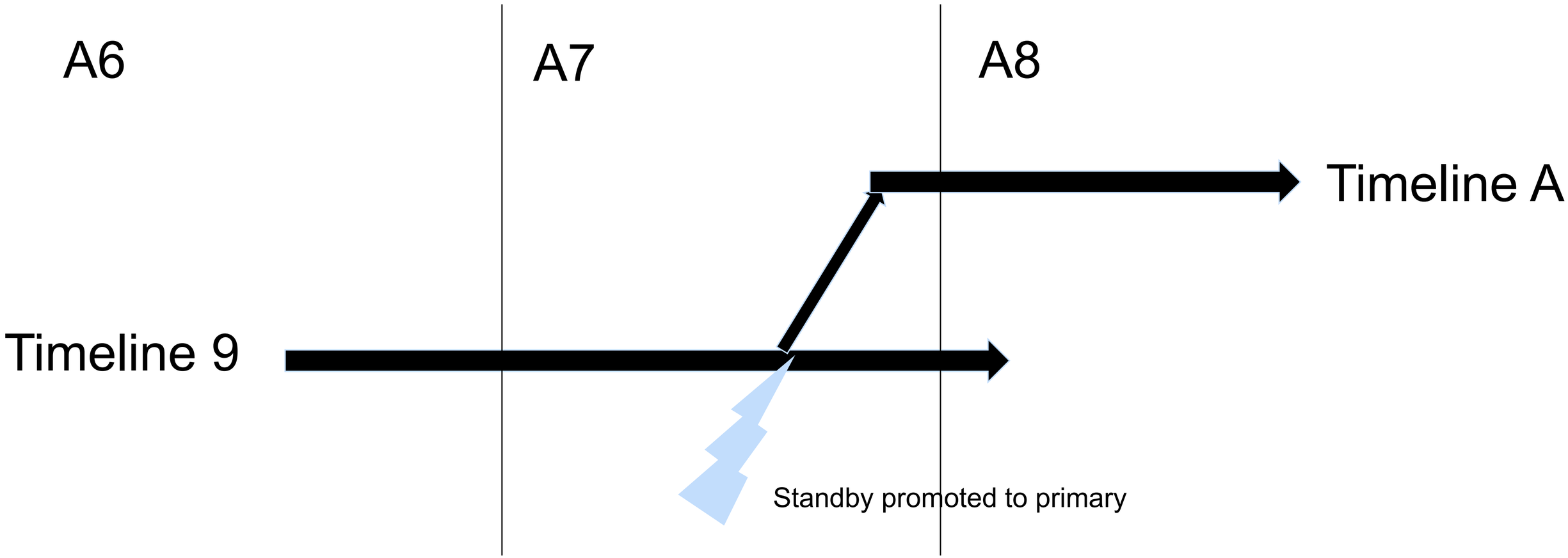
Fix XLogFileReadAnyTLI silently applying divergent WAL from wrong timeline

Edit | Comment/Review | Change Status | Send private mail

ID	6525
Title	Fix XLogFileReadAnyTLI silently applying divergent WAL from wrong timeline
CI (CFBot)	Summary ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✗ ✓ Copy git checkout commands Requeue CFBot
Stats (from CFBot)	Patch version: v2, Patch count: 3, First patch: +242−0, All patches: +418−6
Tags	Bugfix
Created	2026-02-25 08:19:30
Last modified	2026-02-25 08:19:30 (2 weeks ago)
Latest email	2026-02-25 13:19:17 (2 weeks ago)
Status	PG19-Final (2026-03-01 – 2026-03-31): Needs review
Target version	
Authors	Andrey Borodin (x4m)

https://commitfest.postgresql.org/patch/6525/

Switchover



Timeline following

1. "parameter ""primary_conninfo"" changed to ""host=rc1a-skip.mdb.yandexcloud.net""", "postmaster”

2. "WAL receiver process shutdown requested","startup"
"terminating walreceiver process due to administrator command","walreceiver”

3. "started streaming WAL from primary at 2EC2/A7000000 on timeline 9","walreceiver”

4. "replication terminated by primary server","End of WAL reached on timeline 9 at 2EC2/A71F0820.","walreceiver”

5. "fetching timeline history file for timeline 10 from primary server","walreceiver"

Timeline following

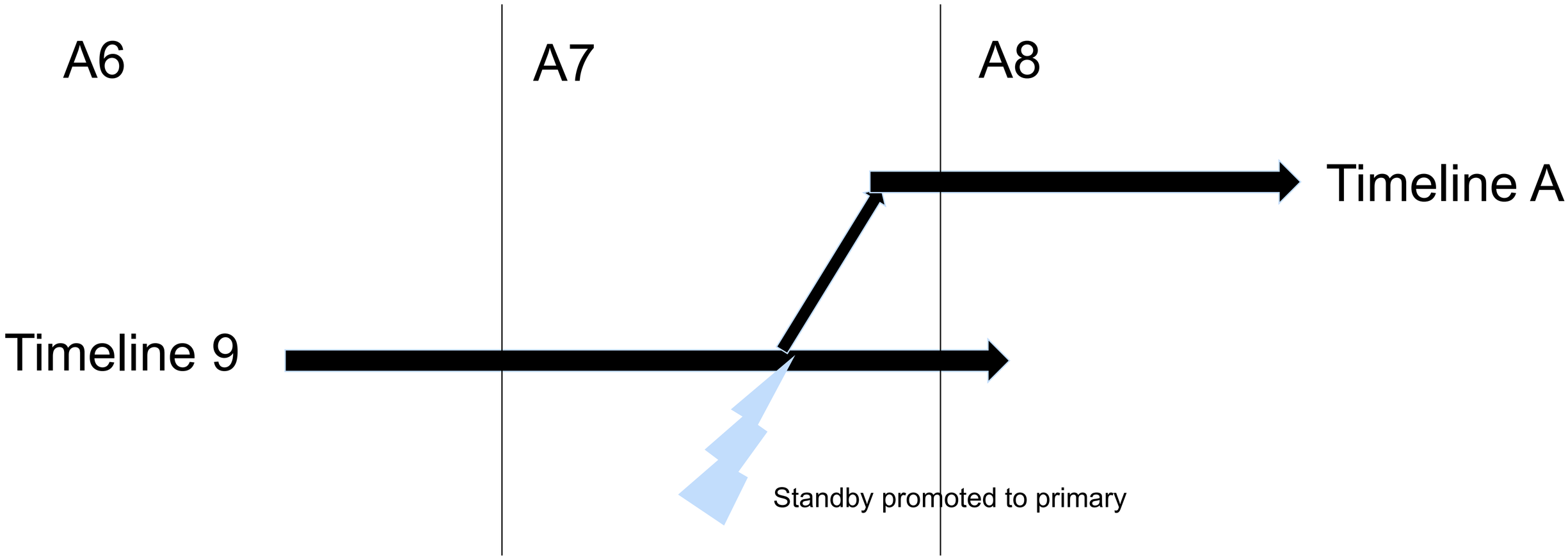
6. "terminating walreceiver process due to administrator command","walreceiver"
"restored log file ""0000000A.history"" from archive","startup”

7. "new target timeline is 10","startup”

8. "restored log file ""00000009000002EC2000000A7"" from archive","startup”

"unexpected pageaddr 2EC2/8000000 in log segment 00000009000002EC2000000A8, offset 0","startup"
"started streaming WAL from primary at 2EC2/A8000000 on timeline 10","walreceiver"
"could not receive data from WAL stream: ERROR: requested starting point 2EC2/A8000000 is ahead of the WAL flush position of this server 2EC2/A71F6AE0","walreceiver"

Switchover



Условия проявления бага

- **Гонка walreceiver и start с SIGTERM в сторону walreceiver**
- **Во время switchover в архиве нет сегмента**

› или

- **Выбранный timeline не достижим в архиве =>**
- **реплика поедет куда попало**

Давайте развивать и чинить Postgres вместе!

Yandex Cloud

PG BootCamp Russia 2026 Moscow
PGBootCamp.ru

Спасибо!

Андрей Бородин

Руководитель подразделения разработки
РСУБД с открытым исходным кодом

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@x4mmm